

The Order-One Nullspace of the Standard Model Finite Spectral Triple

A 46/402 Census

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2026-09-27 — v1.0-draft

Draft for publication. Not peer-reviewed.

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Abstract

We compute the full space of Dirac operators satisfying the order-one condition for the fixed Standard Model finite spectral triple ($A_F = C (+) H (+) M_3(C)$, $H_F = C^{32}$): it is 46 real-dimensional per generation, decomposing into the 10 conventional Yukawa/Majorana parameters, 8 flipped-Yukawa directions, and 28 exotic-Majorana directions. The textbook uniqueness result (Chamseddine–Connes–Marcolli, Theorem 2.21) recovers the 10-parameter form only after imposing the additional $[D, C_F] = 0$ "massless photon" condition — which is baked into that paper's definition of a Dirac operator and removes exactly the 36 extra directions (verified: all 36 fail $[D, C_F] = 0$; all 10 SM directions satisfy it). We extend the census to three generations ($402 = 3 \times 46 + 3 \times 88$ real dimensions, residuals $\sim 1e-15$), classify the 264 generation-mixing directions, and show charge conservation plus one-Higgs minimality selects 84 directions — exactly the Standard Model flavor parameter count. The one-generation order-one theorem is machine-verified in Lean 4 (144/144 generator pairs, zero sorrys). Order-one admits; it does not select.

1. Introduction

The noncommutative-geometric formulation of the Standard Model encodes its internal degrees of freedom in a finite real spectral triple $(A_F, H_F, D_F, J_F, \gamma_F)$ with $A_F = C \oplus H \oplus M_3(C)$ and $H_F = C^{32}$ per generation [1,2]. The Dirac operator D_F carries the Yukawa and Majorana parameters; consistency conditions — the order-zero and order-one conditions — constrain its form. It is widely stated that these conditions select the Standard Model Dirac operator essentially uniquely. We show, by explicit computation, that this is not so: order-one alone admits a 46-real-dimensional space per generation, of which the Standard Model occupies 10 dimensions. The remaining 36 are genuine order-one directions, removed in the literature only by an additional condition that is usually left implicit in the statement of uniqueness.

2. Background and prior art

Chamseddine-Connes-Marcolli [1] classify Dirac operators for the SM finite triple. Their Definition 2.20 states: "A Dirac operator is a selfadjoint operator D in H_F commuting with J_F , C_F , anticommuting with γ_F and fulfilling the order one condition $[[D, a], b^0] = 0$ " — where $C_F = \{(\lambda, \lambda, 0)\}$ is a central subalgebra whose physical meaning (Remark 2.19) is "to ensure that the photon will remain massless." Their Theorem 2.21 ("all Dirac operators are $D(Y)$ ") is proved under this definition. The $[D, C_F] = 0$ requirement is therefore definitional, not derived — and every downstream uniqueness claim inherits it.

Related work: Krajewski [3] classifies abstract finite triples via diagrams; Cacic [4] constructs moduli spaces for fixed multiplicity matrices (the SM diagram yields the 10-dimensional SM space). Boyle-Farnsworth [5] introduce a new second-order constraint precisely to force D_F into SM form — independent evidence that order-one alone does not. Devastato-Lizzi- Martinetti [6] classify the most general Majorana Dirac for an enlarged grand algebra, a different input. Jimenez-Restrepo-Rivera [7] construct a two-Higgs-doublet NCG model by modifying the finite geometry; we exhibit the second doublet's 8 Yukawa slots already sitting in the unmodified triple's order-one nullspace. A 2025 review [8] restates the classification with $[D, C_F] = 0$ still in force.

What is new here. No published work computes the order-one-only nullspace — the intermediate object between "order-one" and "order-one plus $[D, C_F] = 0$." We compute it (46 per generation, 402 for three), decompose it, locate exactly where the literature's extra condition acts, and extend the census to generation mixing. Separately, we give a zero-sorry Lean 4 verification of order-one for the imposed one-generation SM ansatz; no spectral-triple content exists in Physlib, the community Lean physics library, as of September 2026 (verified by source-tree

inspection).

3. One-generation census: 46

Scanning the full 24-real-dimensional generator algebra of $A_F = C (+) H (+) M_3(C)$ in its 32-dimensional representation, the space of selfadjoint gamma-odd D with $[[D, \pi(a)], \pi^0(b)] = 0$ for all 576 generator pairs is 46 real-dimensional (T4 numerical; max residual $1.78e-14$). Decomposition by support and species:

Sector	Real dims	Description
SM Yukawa + Majorana	10	the conventional $D(Y)$ parameters
Flipped Yukawas	8	gamma-odd, J-compatible; "wrong-Higgs" slots
Exotic Majorana	28	14 complex; (ν_R, ν_R) -type and charged slots

Structural invariants: the C -block vanishes identically; $B = \text{conj}(A)$; all 36 extras are grading-odd, J-compatible, self-adjoint — genuine order-one directions, not numerical artifacts. Two measurement artifacts were caught and corrected by independent verification scripts during the census (recorded in the repository, not laundered).

4. Where the literature's extra condition acts

For each of the 46 directions D_k we computed $||[D_k, P_{\{H(+)C\}}]||$, where $P_{\{H(+)C\}}$ represents the central subalgebra $C_F = \{(\lambda, \lambda, 0)\}$:

- All 10 SM directions: $||[D, C_F]|| = 0$ exactly. - All 36 extra directions: $||[D, C_F]|| \geq 0.48$ — none survive.

Hence the $36 = \ker([., C_F])^\perp$ intersected with the order-one nullspace, exactly. This refines our earlier "27/28 killed by charge conservation" phrasing: electric charge was a proxy that mislabeled the mechanism. The precise killer — including for the EM-neutral (ν_R, ν_R) direction — is $[D, C_F] = 0$. The CCM proof invokes C_F at exactly the two places our decomposition predicts: cutting the 16-dimensional S -sector to the 8 SM Yukawas (our 8 flipped), and forcing the Majorana T -block to the single (ν_R, ν_R) slot (our 28 exotic).

5. Three generations: 402

Imposing $H_F = C^{32} \times C^3$ with arbitrary 3×3 complex Yukawa matrices and symmetric 3×3 Majorana matrix (triplication imposed, not derived): the admissible Dirac space is 2352 real-dimensional; the order-one nullity is $402 = 3 \times 46$ (diagonal) + 3×88 (mixing), verified over random trials (residuals $\sim 1e-15$, non-vacuous probes). Per generation pair, the 88 mixing dimensions decompose into slot-disjoint subspaces:

Family	Real dims	Fate
SM-like mixing	18	survives: off-diagonal flavor entries (CKM/PMNS-type)
Flipped Yukawa mixing	16	shelved (one-Higgs minimality)
Exotic Majorana mixing	54	killed by $[D, C_F] = 0$

No genuinely new structures appear beyond the one-generation verdict crossed with 3×3 flavor (one structural change: E-symmetry drops in the mixing sector).

6. Selection: 84

Charge conservation ($[D, C_F] = 0$) plus one-Higgs minimality selects 3×10 (diagonal) + 3×18 (mixing) = 84 directions = 4×18 (four 3×3 complex Yukawa matrices) + 12 (symmetric 3×3 M_R) — exactly the Standard Model flavor parameter count. Order-one admits 402; the selection is physical input; the count matches.

7. Spectral-action cross-check

Over the selected 84-direction space: $\text{Tr}(D^2) = 4(\|Y_{\nu}\|^2 + \|Y_e\|^2 + 3\|Y_u\|^2 + 3\|Y_d\|^2) + 2\|M_R\|^2$ and $\text{Tr}(D^4)$ as computed in [9]; the brackets reproduce the Chamseddine–Connes–Marcolli coefficients a, b, c, d, e structurally (inner fluctuations off; cutoff moments kept symbolic — no Higgs-mass or cosmological-constant prediction claimed).

8. Lean 4 verification

MartinettiRep.lean: order-zero for 144/144 selected generator pairs; $J^2 = 1$; $J\gamma = -\gamma J$; full unitality; and the full order-one theorem for the imposed one-generation Dirac ansatz over arbitrary complex Yukawa/Majorana parameters — all 144 pairs, zero sorrys, full lake build (3,316 jobs) green. Scope, honestly: the Lean theorem covers the imposed SM ansatz over the selected 12-generator family, not the full 46-dim space; the ansatz is input, not derived. The three-generation Lean scaffolding (ThreeGen.lean) defines the tripled operators; the diagonal three-generation order-one reduction is deferred.

9. What this does and does not show

Does show: the order-one condition, stated alone, admits a precisely enumerated 46/402-dimensional space; the published uniqueness theorem is uniqueness modulo $[D, C_F] = 0$, and we exhibit the 36 directions that condition removes; the selected 84 match the SM flavor count; the one-generation order-one theorem is machine-checked.

Does not show: a derivation of the Standard Model (the ansatz and the selection are inputs); a derivation of three generations (imposed); a derivation of flavor structure (CKM/PMNS not derived); any mass-ratio prediction (permanently quarantined: no legitimate pre-registerable series exists in the current axioms).

10. Conclusion

The order-one condition is weaker than the literature's working definition of "Dirac operator," and the gap is now measured: 36 directions per generation, 318 for three, each named, each residual-audited. Nothing in the published mathematics is contradicted; something the published mathematics never computed is now on the record — with a machine-checked core and a permanent, labeled boundary around what is input.

Data and code: github.com/jeffgurd888/thet-logos (attack4_corrected/: DIRECTIONS_36.md, THREE_GEN.md, MIXING_264.md, SPECTRAL_ACTION_84.md; lean/ThetLogos/: MartinettiRep.lean, ThreeGen.lean).

References

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